

Measurability of the calibration entries

Direct statistical counterparts, indirect constructions, and latent model quantities

This document assesses **every one of the 35 grouped entries** in `V6_calibration_core.tex`, in the same order. It uses the V6 equations, the zero-exponent solver, the earlier `Codes/Empirical_Data` mappings, the FinAI measurement workflow, and fresh WRDS/OpenEcon/official-API checks. The classification concerns an *accurate model-to-data mapping*, not simply whether a similarly named series exists.

Read the classification together with mapping readiness

Class	Meaning
D: directly measured	A specified statistical counterpart is reported or constructed by transparent aggregation, ratios or unit conversion, without imposing model behavioral equations. Sampling error and ordinary national-accounts estimation do not disqualify D. D[†] means the model correspondence is conditional: the population, sector, instrument or timing convention still needs to be adopted and reconciled.
I: indirect / model implied	A missing model coordinate requires a substantive bridge, a proxy, a normalization or a model inversion. The table distinguishes these cases. I does <i>not</i> certify that a weak proxy is accurate, and a model-generated value is not independent empirical evidence.
L: structurally latent	No verified observation rule measures the preference, technology or regime construct. It requires joint structural identification, a compatible external estimate, or an explicitly imposed restriction. L does not mean dispensable or inherently unidentifiable.

Assessment. Five entry families have direct statistical candidates, all conditional; 13 require an indirect/model-implied construction; 17 are structurally latent, including the imposed zero exponents and the unmapped regime. Country/component qualifications remain visible: for example, US wage observations do not supply missing RoW wage levels. **No conditional D entry is being declared calibration-ready.** Under a strict requirement of an already completed model correspondence, those conditional D entries remain unresolved bridges too.

The most promising direct anchors are explicitly defined employment/earnings, output accounts, gross external equity positions and NFA. The principal unresolved mappings are research labor within skilled labor, knowledge varieties, factor augmentation, complete resident equity portfolios, the single safe bond, and the relation of enterprise value/FCF to model equity/profits.

Verification is separate from identification. Live WRDS samples, 19 exact FRED series and two exact Eurostat samples establish access to specified observations. They do not establish a full historical panel or an identified calibration. Both OpenEcon checks returned a different concept from the one requested; their substitutions were rejected. The source column links to the evidence ledger and distinguishes live observations, current metadata and previously saved panels.

1 Parameters: saving, portfolios and regime risk

Core entry	Class / readiness	Proposed mapping and accuracy judgment	Data source / verification
M01 β	L Structural identification	No accurate direct mapping. BEA personal saving/disposable income is aggregate net saving; $\beta = A/e_{US}$ concerns cohort asset acquisition/labor income, including purchases from the old. SCF wealth/income is not that flow either. Identify jointly after choosing the OLG period and the asset/income boundary.	S1 BEA NIPA saving; SCF assets, income and saving questions. Metadata verified; contextual evidence only.
M02 γ	L Structural identification	Risk aversion in the specified continuation utility has no data column. Risk-attitude answers and realized returns can discipline a structural preference estimate, but neither equals this curvature. External estimates require a compatible utility specification and population.	S1, F2 SCF questionnaire; CRSP returns via WRDS. Survey metadata and live return sample.
M03 κ	L Structural identification	Portfolio data discipline a return-wedge/target-deviation combination, not this preference curvature alone. A single equity share cannot separate κ from $\bar{\omega}$, risk preferences and expectations. Requires allocation responses or explicitly imposed restrictions.	F2, F3 CRSP returns; resident equity holdings. Neither source directly measures utility costs.
M04 $\bar{\omega}, \bar{\omega}^*$	L Structural identification	These are preference centers, not realized home bias. Observed US-equity portfolio weights identify the actual allocation only. Setting the centers equal to observed weights is an additional restriction on risk-adjusted return wedges, not direct measurement.	F3 Fed international equity positions; domestic equity accounts needed for denominators. TIC/IMF PIP supplementary.
M05 χ	L Structural identification	A safe-asset spread or foreign Treasury share does not equal convenience utility. The model links χ jointly to holdings, marginal valuation and the safe-rate wedge; it also changes RoW saving. Currency, credit and duration differences in observed bonds must be separated from convenience.	F6, S1 H.15/TB3MS and Treasury TIC holdings; household saving evidence. No exact χ series.
M06 η	L Structural identification	The coefficient scales a utility cost of the bond share, not observed underwriting fees or a resource cost. It needs joint identification from holdings and risk-adjusted return wedges. $\Upsilon(\theta) = -\log(1 - \theta) + \theta^2/2$ is imposed by the code.	F2, F6, M1 Equity returns, bond yields/positions; model first-order condition. Functional form is fixed.
M07 π	L Regime definition missing	Persistence becomes estimable only after an operational definition of u/b and a period length. A recession indicator or a chosen 2008 date is not the model technology regime. An all- u episode gives censored duration evidence, not a directly observed interior transition probability.	M1, P5 No verified regime series. Productivity/investment histories are possible inputs to a future state rule.

2 Parameters: production and knowledge

Core entry	Class / readiness	Proposed mapping and accuracy judgment	Data source / verification
M08 H_i, L_i	D[†] Conditional: scope and units	Direct employment counts/hours exist for declared skill groups. Candidate: US CPS BA+ versus exhaustive non-BA; Eurostat tertiary versus non-tertiary, age 25–64. Adopt a common age, sector, headcount/FTE and country convention. Education as model skill is a maintained bridge; efficiency units remain I.	P1 CPS ORG: local micro mapping. Eurostat 1f5a_egaed: live DE sample, annual thousand persons.
M09 a_i	L Structural identification	$a_i = (N_{i,t+1}/N_{i,t} - 1)/[(1 - \varphi_i)H_i]$ is conditional recovery, not direct measurement: neither knowledge varieties nor matched research labor is yet observed. Free-entry inversion uses the same valuation bridge. Separating a_i from $a_i H_i$ needs the labor unit and model period.	P2, P3, P1 MSTI/BERD research inputs; R&D/patent stocks; labor quantities. Existing implied- a series is a diagnostic.
M10 ϑ_i	L Structural identification	The inverse monopoly markup is not an accounting gross margin or dividend share. An inversion using $w_H \varphi H$ and model profit $\mathcal{D}$ requires the same production sector and treatment of physical capital, fixed costs, taxes and retention. Firm accounts supply ingredients, not the markup itself.	F1, P4, P1 Compustat expenses/sales; corporate accounts; wages. WRDS access confirmed; no direct markup field.
M11 α_i	L Normalization and identification	Not an observed labor-income share. CES levels depend on $\alpha_i A_{X,i}^{1-\rho_i}$ and $(1 - \alpha_i) A_{L,i}^{1-\rho_i}$. Separating distribution weights from augmentation scales requires normalization and structural restrictions; downloading factor shares does not resolve this.	P1, P4 Employment, wages and corporate accounts are joint calibration evidence.
M12 ρ_i	L Structural identification	The elasticity is $1/\rho_i$. Identification needs relative-input/price variation and assumptions on markup and augmentation dynamics. A generic capital–labor elasticity is not automatically the intermediate-input/unskilled-labor elasticity used here. Both countries use CES in the solver.	P1, P4, M1 Labor quantities/prices; production accounts; compatible external structural estimates.
M13 $\bar{A}_{X,US,u}, \bar{A}_{L,US,u}$	L Structural identification	Factor-specific scale coefficients need a production inversion conditional on α, ρ, ϑ , knowledge normalization and output/input units. R&D spending, sales per employee, and industry TFP do not measure these coefficients.	P1, P4, P5, F1 Labor/production accounts and firm R&D are ingredients. No verified direct augmentation series.
M14 $\bar{A}_{X,US,b}, \bar{A}_{L,US,b}$	L Counterfactual restriction	Require measured absorbing-regime production with a justified state rule, or a counterfactual calibration restriction. An all- u sample does not reveal absorbing productivity. Labeling a historical crash as b does not establish the technology correspondence.	P4, M1 Production histories plus an independently specified regime rule; currently no exact branch mapping.
M15 $\bar{A}_{X,W}, \bar{A}_{L,W}$	L Structural identification	RoW augmentation scales need harmonized quantities, wages and output under the solver CES. A sector productivity ratio or a weighted country average is not either augmentation. Covered G4/OECD-ex-US panels are not the entire residual world.	P1, P4, P5 Eurostat/OECD labor and institutional accounts. Matched RoW wage levels and scope remain gaps.
M16 ξ_u, ν_u	L Structural identification	Elasticities of separately inferred augmentations with respect to a separately mapped knowledge stock. Relative wages alone cannot identify both. Defining $A_X = N^{\xi_u}$ and comparing that generated series to N imposes the elasticity rather than measuring it.	P3, P1, P5 Knowledge proxies and production evidence; no directly measured elasticity pair.
M17 $\nu_b = 0, \xi_W = 0$	L Imposed, not estimated	Known by construction in the chosen solver; not measurements of zero elasticities. Retain as maintained restrictions and test implications. Positive R&D investment growth does not by itself reject zero factor augmentation. Reopening these coefficients changes the computational specification.	M1, P5 validate_params and zero-exponent restriction. R&D investment is only contextual evidence.

3 Exogenous inputs

Core entry	Class / readiness	Proposed mapping and accuracy judgment	Data source / verification
M18 $N_{US,0}, N_{W,0}$	I Normalization / proxy	Setting initial knowledge to one is a normalization, not a measured level. A patent or R&D-capital proxy requires office/residence, timing, depreciation and initial-stock conventions. Rebase only after these are fixed; normalized units do not remove the proxy gap.	P3 Fed/BEA R&D stocks; USPTO/OECD/WIPO patent alternatives in the local package.
M19 $z_t \in \{u, b\}$	L Empirical state rule missing	Exogenous in the model does not mean observed in data. A switch date τ imposed in a simulation is a scenario. A documented empirical regime classifier would yield an inferred state (I); no accurate operational u/b mapping is presently supplied.	M1 Code path available; no verified empirical state series.

4 Endogenous states and equilibrium choices

Core entry	Class / readiness	Proposed mapping and accuracy judgment	Data source / verification
M20 $N_{i,t}$	I Model state / proxy	The law of motion gives a model-implied path. R&D-cost stocks and perpetual-inventory patent stocks are observable constructions but do not count model varieties. Their growth depends on valuation, depreciation, office coverage and initialization; preserve distinct proxy families.	P3 Q4 R&D stocks FL105013465, FL795013465; patent stock alternatives. Local evidence.
M21 $\varphi_{US,t}, \varphi_{W,t}$	I Research/skill bridge missing	Need $\varphi = 1 - R/H$ with research hours/persons a subset of the same skilled population. $NRC \cap H/H$ fixes the denominator but remains a task proxy: management/clinical work is not necessarily innovation. BERD/MSTI researcher or personnel FTE and headcount are different definitions.	P1, P2 CPS/NRC and BERD/MSTI. Live OpenEcon researcher response rejected for wrong measure; OECD HTTP 403 retained.
M22 ω_t, ω_t^*	I Portfolio-universe bridge	Construct actual US-equity weights from residence-consistent domestic and foreign equity holdings; both denominators include all equity in the same portfolio. External positions alone are insufficient. A foreign ownership share of US capitalization is not ω^* . Match equity/FDI/fund coverage and consolidate intermediaries.	F3, F7 Fed cross-border equity plus domestic equity-supply accounts. Feasible accounting recovery is given on page 7.
M23 $\theta_{US,t}^*$; recovered $\theta_t, \theta_{W,t}^*$	I Safe claim and wealth gap	Foreign US-safe-claim holdings need the corresponding investable-wealth denominator. Treasury holdings span maturities and private/official institutions. Neither debt/GDP nor all non-equity IIP is this share. $\theta_W^* = 0$ and $\theta = -\theta_{US}^* A^*/A$ are equilibrium reconstructions, not empirical observations.	F6, F4, M1 Treasury TIC; ROWTSEQ027S; model saving/clearing relations.
M24 $R_{f,t}, R_{f,t}^W$	I Period/numeraire bridge	A named nominal bill yield is D, but TB3MS is a monthly average of annualized discount-basis quotes. Match the holding period and payoff convention. Ex-post inflation adjustment does not create an ex-ante safe real payoff. The zero-supply RoW bond has no uniquely observed world issuer/currency; its yield is model implied.	F6, M1 Fed H.15 TB3MS: live through Aug 2026. RoW model shadow yield has no exact series.

5 Endogenous production and valuation

Core entry	Class / readiness	Proposed mapping and accuracy judgment	Data source / verification
M25 $A_{X,i,t}, A_{L,i,t}$	I Conditional model inversion	Can be inferred only conditional on CES parameters and matched production inputs/prices. That inversion is model implied and cannot independently validate the same parameters. Sector labor productivity, R&D GFCF and knowledge-cost deflators are not factor augmentations.	P1, P4, P5 Labor and accounts supply inversion inputs. Live R&D-investment observations establish expenditure, not augmentation.
M26 $Y_{i,t}$	D[†] Conditional: output definition	Corporate GVA or a declared output aggregate is directly measured statistically. Choose its sector, real-price basis and country coverage; GDP and corporate GVA are not interchangeable. Reconcile national-accounts capitalization of R&D with model $Y = e + \mathcal{D} - \mathcal{I}$ before adopting GVA as model output.	P4, P5 US IMA corporate GVA; OECD $B1G[S11] + B1G[S12]$. Existing mapped panel and verified official definitions.
M27 $w_{H,i,t}, w_{L,i,t}$	D[†] Conditional; RoW levels gap	US mean earnings per compatible labor unit and skill group are direct candidates. Earnings exclude benefits; median weekly wages are not mean hourly compensation. Corporate-compensation scaling is I. Current RoW relative-earnings panels do not supply matched wage levels, so that component remains I/gap.	P1, P4 US CPS ORG micro wages; BLS wage sensitivities. OECD EAG ratios and Eurostat labor quantities: local mappings.
M28 $e_{i,t}$	I Income/coverage bridge	Construct $w_H H + w_L L$ using matched workers, wages and time units. Aggregate corporate compensation is a candidate accounting bridge, not automatically young-cohort opportunity income. Forcing micro wages to reproduce compensation is an identity by construction, not independent validation. RoW wage ratios alone do not recover income levels.	P1, P4 US corporate compensation $FU106025005 + FU796025005$; micro wage-bill alternative.
M29 $q_{i,t}, d_{i,t}$; reconstructed $R_{i,t+1}$	I Variety-unit / price bridge	Per-variety levels need $q_t = Q_t/N_{t+1}$ and $d_t = \mathcal{D}_t/N_t$, hence the knowledge mapping. Named traded-security price/total returns are D candidates for growth moments only. Splits, issuance and changing composition prevent raw share prices or capitalization growth from directly measuring q .	F2, P3, F5 CRSP <code>prc,ret,retx</code> ; live 2024 sample. Earlier AHP revaluation-implied price chain requires coverage reconciliation.
M30 $\mathcal{Q}_{i,t}, \mathcal{D}_{i,t}$	I Claim/cash-flow bridge	Equity capitalization and consolidated distributions can be D for a specified universe. Current AHP enterprise-value/FCF candidates are I for V6: enterprise value is not shareholder equity, and FCF is not paid dividends or the model profit flow. Retention, repurchases, taxes, debt and intercorporate payouts require reconciliation.	F7, F1, P4 Fed EV and dividend alternatives; OECD F51 liabilities; Compustat accounts. Keep each mapping ID separate.

6 External positions, valuation and flows

Core entry	Class / readiness	Proposed mapping and accuracy judgment	Data source / verification
M31 $qW_{t,t}n_{W,t}, qUS_{t,t}n_{US,t}^*$	D[†] Conditional: equity universe	Residence-based market-value external equity positions are direct counterparts. Specify FDI, fund shares and international-organization equity; current asset/liability source universes differ in fund coverage. Match timing and currency. The underlying model claim counts n remain I after choosing a price normalization.	F3 ROWEISQ027S, ROWEINQ027S; live exact series through 2026Q2; Fed analyzer definitions checked.
M32 $b_t = \theta_t A_t$	I Safe-bond aggregation gap	$NFA - E_A + E_L$ is a measurable non-equity residual, not automatically one safe bond. It includes loans, deposits, securities, derivatives and reserves in broad accounts. Foreign Treasury holdings are gross liabilities, whereas b is a consolidated net present-value position. Specify the claim and netting rule.	F3, F4, F6 International total/equity accounts; TIC Treasury holdings. Broad residual is descriptive, not a direct safe-bond measure.
M33 NFA_t	D[†] Conditional: instrument scope	Whole-economy NFA is directly measured by external assets minus liabilities; transparent subtraction does not make it latent. To match V6, choose a consistent instrument universe or reconcile omitted assets/claims in an explicit accounting bridge. Do not infer observed cohort wealth merely by imposing $A = NFA + Q_{US}$.	F4 ROWTLEQ027S-ROWTASQ027S=-ROWNETQ027S. Exact identity in all 10 sampled quarters.
M34 $VA_t^{asset}, VA_t^{liability}, VA_t$	I Price/valuation bridge	Published revaluations are direct statistical inputs, but model $n_{t-1}\Delta q_t$ requires matched holdings and pure price changes. Separate FX/other changes and preserve the negative US-liability sign. The legacy liability price chain pairs a broad stock with a narrower revaluation series; resolve that difference first.	F5, F3 Asset FR263181105; liability FR263081005 versus FR263081115. Live coverage/value discrepancy verified.
M35 $CA_t, \Delta NFA_t$	I Transactions/closure bridge	Observed ΔNFA and reported CA are D statistical quantities. The current model mapping remains I because it assigns all Δb to transactions. Empirical bond valuation, capital-account/statistical discrepancies and other changes must remain separate; do not force $\Delta NFA = CA + VA$ by deleting residuals.	F4, F5 Quarterly US CA candidate $-RWBACQ027S/4$; NFA levels and separate FU/FR accounts. Annualization verified.

7 Useful measurement constructions and their limits

Actual equity weights can be constructed without treating preference centers as data. Let $E_A = q_W n_W$ and $E_L = q_{US} n_{US}^*$ be measured external equity positions. If domestic equity supplies Q_{US}, Q_W use the same equity/FDI/fund universe, residence rule and consolidation, then

$$\omega = \frac{Q_{US} - E_L}{Q_{US} - E_L + E_A}, \quad \omega^* = \frac{E_L}{Q_W - E_A + E_L}.$$

These are feasible accounting constructions for M22, not observations of $\bar{\omega}, \bar{\omega}^*$. Operating-asset enterprise value cannot silently replace equity supply; covered OECD countries cannot silently replace the entire RoW. National end-date holdings may be the chosen aggregate counterpart of model investors without literal age filtering, but that does not equate national net saving with young-cohort asset acquisition.

Knowledge and augmentation require separate, explicit bridges. The identities $q_{i,t} = Q_{i,t}/N_{i,t+1}$ and $d_{i,t} = D_{i,t}/N_{i,t}$ preserve V6 timing. For $X_i = \varphi_i H_i$ and $\rho_i \neq 1$, a conditional CES inversion is

$$A_{X,i} = \left[\frac{\vartheta_i \alpha_i Y_i^{\rho_i}}{w_{H,i} X_i^{\rho_i}} \right]^{1/(\rho_i-1)}, \quad A_{L,i} = \left[\frac{(1 - \alpha_i) Y_i^{\rho_i}}{w_{L,i} L_i^{\rho_i}} \right]^{1/(\rho_i-1)}.$$

This requires the production accounting check $w_H X/\vartheta + w_L L = Y$, compatible input/wage units and all the indicated parameters. It is undefined at $\rho = 1$ and sensitive nearby. Computing it does not independently identify or validate those parameters. Do not rescale inconsistent empirical factor payments merely to enforce the equality.

The current external-account bridges need a coverage reconciliation. The earlier liability price chain combines ROWEQ027S/FL263081005 with FR263081115. Current Fed component definitions link the broad stock to FR263081005, while FR263081115 excludes mutual/money-market fund shares. In 2026Q2 the live values are 3,921,264 and 3,718,464 million USD: a **202,800 million USD difference**. This establishes a scope difference, not a license to replace one series automatically. Reconcile levels, transactions, revaluations and other changes together. See F3 and F5.

For M35, keep empirical CA, net financial transactions, equity VA, FX/bond VA, other changes and statistical discrepancies distinct. The model's assignment of all Δb to CA is not valid for arbitrary long-maturity market-value debt. Notebook 06's window statistics divide cumulative flows and the change in the NFA *level* by end-window output; they are not sums of flow/output ratios or a change in NFA/output. Select the sectoral output denominator and model-period frequency before constructing calibration moments.

8 What the live checks established

Route / sample	Verified result and interpretation
WRDS: Compustat NA	<code>comp.funda</code> , gvkey 001690/fyear 2024 returned 2 formats. Exactly 1 row is C/INDL/STD; the other is SUMM_STD. R&D, employee, payout and cash-flow fields exist in the standard row. This confirms a firm-level route, not national research/skill quantities. Duplicate-format summation would be wrong.
WRDS: Compustat Global	<code>comp.g_funda</code> , fyear 2024 returned an intentionally capped 3-row pilot; 42,376 matching rows were advertised. The pilot is incomplete and not representative. No complete Global panel, growth trend or aggregate moment was inferred.
WRDS: CRSP	<code>crsp.msf</code> , PERMNO 14593 returned all 12 unique months of 2024 with price, distribution-inclusive/exclusive return and adjustment fields. The provider count reported 3 despite 12 returned records. Calendar/identifier validation established the bounded sample; the inconsistent metadata were preserved. This does not certify coverage of all securities or a current national index.
Exact FRED / Fed data	19 exact series: 16 quarterly through 2026Q2, TB3MS through Aug 2026 and 2 annual dividend series through 2025. The request window began in 2024. Dates are unique/sorted; the NFA identity holds exactly in all 10 sampled quarters. These are saved-sample dates, not a claim about full source histories.
Exact Eurostat: labor	<code>lfsa_egaed</code> , DE, age 25–64, both sexes, ED5-8, THS_PER: 12,543.1; 12,708.1; 13,052.0 thousand employed persons in 2021–23. The 2021 break flag is preserved. This directly measures tertiary employment, not corporate skill efficiency or research labor.
Exact Eurostat: R&D	<code>nama_10_a64_p5</code> , DE, TOTAL, N1171G, P51G, CP_MNAC: 105,966; 110,894; 124,513 million national currency in 2021–23. The 2022–23 provisional flags are preserved. This is an investment flow, not knowledge varieties, innovation success or A_X .
OpenEcon: two checks	Requested exact ROWEISQ027S, received FL263164100Q: the 2026Q2 values differ by 11,305,117 million USD. Requested MSTI business researchers B_RS/FTE, received G+T_RS URLs and generic units. Both responses contained numbers; both were rejected for these mappings. Direct exact-source probes supplied the usable evidence.
Failed routes retained	The exact OECD MSTI direct request returned HTTP 403; earlier saved OECD data remain separately labeled. Initial network/time-out routes and unsuccessful candidate endpoints are logged. Failure of a route is not evidence that the economic series does not exist.

Reproducible package. `measurability_evidence/` contains immutable-for-this-audit raw samples, request/response receipts, source IDs/URLs, flags, validation summaries and the classification CSV/JSON. WRDS metadata are in `wrds/tool_receipts.json`; official probes and rerun scripts are under `production/` and `external/`. The PDF is rebuilt by `build_tables.py` plus LaTeX. A rerun can receive revised source data; retain this dated snapshot when changing vintages. No model solve, parameter estimation, bulk historical refresh or change to the earlier mappings was performed.

9 Data-source and verification ledger

Source keys used in the tables are defined below. Live sample dates identify the data actually retrieved; local panels retain their earlier vintage. URLs point to the producer or authenticated data endpoint. The accompanying source ledger records the full descriptions.

S1. Household saving and preferences

Official metadata checked; no new SCF microdata.

BEA defines personal saving relative to disposable personal income; SCF collects family assets, debts, income and attitudes. These observations do not directly identify the model utility coefficients.

BEA saving definition; SCF public data; SCF questionnaire.

P1. Skill labor and wages

Local CPS/EAG panels; live Eurostat sample and metadata.

US CPS ORG local baseline: private-for-profit employees, 1994–2024, BA+ versus exhaustive non-BA, headcount or matched-hours FTE. Eurostat `lfsa_egaed`: annual thousand persons, sex T, age Y25-64, ED5-8/ED0-2/ED3_4. Local RoW earnings ratios do not establish wage levels. BLS definitions; LFS metadata; Exact DE API sample.

P2. Research labor

Local official data/primary metadata; live OECD route 403.

OECD MSTI v1.3: business researchers B_RS or total personnel B_TT, FTE versus persons. NCSES BERD 2023 Tables 50/52 cover domestic business employment/research personnel; current local R&D FTE panel is 2015–2023. Survey scope and numerator membership in H must match.

OECD MSTI; BERD Table 52.

P3. Knowledge proxies

Existing frozen mappings; no exact-variety observation.

Fed/BEA Q4 R&D net stocks FL105013465+FL795013465; investment-price deflator Y006RG3Q086SBEA is a sensitivity, not an official stock-volume index. USPTO grants, OECD triadic and WIPO applicant-residence stocks have distinct coverage, initialization and depreciation.

NFC R&D stock; Financial R&D stock.

P4. Corporate output, labor income and equity

Existing 2026-09 mapping vintage; producer definitions checked.

US GVA: FU106902501Q+FU796902505Q; compensation: FU106025005Q+FU796025005Q. OECD institutional accounts: B1G and D1 for S11+S12; F51 liabilities are a distinct equity alternative. Native currencies, market FX/PPP and end-period/flow timing are separate choices.

US corporate GVA component; OECD accounts.

P5. R&D investment and accounting

Live Eurostat data; BEA primary definitions.

Eurostat `nama_10_a64_p5`: N1171G asset, P51G transaction, TOTAL industry; CP_MNAC is current-price million national currency. BEA capitalizes R&D; expenditure and GDP contributions are not innovation success or factor augmentation.

Exact DE R&D API sample; BEA capitalization.

F1. WRDS / Compustat firm accounts

Live OPTIONS and data, 2026-09-22.

`comp.funda`: gvkey 001690/fyear 2024, 2 raw format variants, 1 C/INDL/STD row after local selection. `comp.g_funda`: intentionally capped 3-row 2024 pilot, not a complete or representative sample. Employee/R&D/accounting fields do not give research/skill hours or model markups. API types alone do not certify units.

NA API; Global API; Producer/access overview.

F2. WRDS / CRSP security prices and returns

Live 12-month 2024 sample; metadata discrepancy retained.

`crsp.msrf`, PERMNO 14593: prc,ret,retx,shrout,cfacpr,cfacshr. All 12 unique 2024 months returned. Source count said 3 despite 12 records; completeness was checked against the requested calendar. Security returns are observable, but do not identify a model variety or a national investor portfolio.

Monthly stock API; CRSP product description.

F3. International equity positions

Live exact FRED samples and Fed component definitions.

US external equity assets: ROWEISQ027S/FL263181105; US equity liabilities:

ROWEINQ027S/FL263081005. Quarterly end-period million USD. Current fund/FDI/international-organization coverage must be read from the component trees, not inferred from an old shorthand label.

Asset component tree; Liability component tree.

F4. NFA, transactions and current account

Live samples 2024Q1–2026Q2.

US NFA=ROWTLEQ027S-ROWTASQ027S=-ROWNETQ027S, million USD. US CA candidate=-RWLBACQ027S/4: the source is ROW net lending/borrowing (B9) at an annual rate. Capital-account/statistical items must be reconciled before identifying this candidate with CA. RWNEOWQ027S is ROW net-worth revaluation (US total VA is its negative); equity FU components are transactions. Do not conflate SAAR and period flows.

ROW net worth; ROW net lending.

F5. Equity revaluation and price/FX separation

Live scope discrepancy; current bridge unresolved.

BOGZ1FR263181105Q, BOGZ1FR263081005Q and BOGZ1FR263081115Q are quarterly revaluations in million USD, not annualized transaction flows. BEA Concepts and Methods separates transactions, price/FX revaluation and other changes. Matching FR/FU/stock universes is required.

Broad liability FR; Legacy liability FR; BEA 2026 methods, chs. 8/24.

F6. Safe yields and Treasury holdings

Live series; Treasury coverage notes checked.

TB3MS: monthly average 3-month bill rate, percent at an annual discount-basis quote; saved through Aug 2026. ROWTSEQ027S: quarterly foreign Treasury holdings. TIC Table 5 includes multiple maturities and custody-based country attribution; it is not a model wealth denominator.

TB3MS; TIC Table 5.

F7. Capitalization, operating-asset value and distributions

Live Fed alternatives; local AHP construction retained.

AHP operating-asset values: LM102010405Q+LM792010405Q. Gross-dividend alternatives: FU106121001A+FU796121001A, sampled through 2025. FCF uses operating surplus minus taxes/investment. These distinct claim/cash-flow concepts are not interchangeable; gross dividends also need intercorporate consolidation.

Operating-asset value; NFC dividends.

M1. Model, prior crosswalk and workflow

Local source review; no baseline changes or new estimation.

V6_calibration_core.tex; V6 manuscript; TwoCountryProductionOLG.jl; annual empirical mapping registry and method notes; FinAI AGENTS.md, SMU_RESEARCH and SMU_PROVIDERS. Source authority, access success and economic mapping are separate assessments.